

The consequences of climate change in Atlantic Forest birds

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Abstract

Here we present a preregistration for a phylogenetic comparative study of the effects of climate change on morphology of Atlantic Forest birds. Below we will provide detailed information on the planned research project.

Plain Language Summary

Earthquake data for the island of La Palma from the September 2021 eruption is found ...

1 Introduction

A well-known ecogeographical pattern describes the relationship between environmental temperature variation and body size. This association, known as Bergmann's rule (Bergmann, 1847), states that animals in warmer climates tend to be smaller in size. The logic underlying this rule is that the increase in body surface area to volume ratio should facilitate more efficient thermoregulation (Mayr, 1956). Researchers have found empirical evidence supporting Bergmann's rule both within and across species, specifically for birds (Ashton, 2002; Fr hlich et al., 2023; McQueen et al., 2022). The relationship between warmer temperatures and smaller body sizes has led researchers for decades to predict that anthropogenic climate change may cause changes in body size in a analogous way as the ecogeographical pattern. That is, as the Earth's climate warms, body size in endothermic animals, such as birds, should decrease (Gardner et al., 2011; Sheridan & Bickford, 2011; Celine Teplitsky & Millien, 2014).

Two of the main responses of organisms to climate change are phenological changes and geographical range shifts (Parmesan & Yohe, 2003). Yet, the shrinkage of bodies, a third response of organisms to global warming, is increasingly being observed (Gardner et al., 2011; Sheridan & Bickford, 2011; Celine Teplitsky & Millien, 2014). Body size shrinkage due to climate change has been hypothesized to be a consequence of ecological or evolutionary processes (Ozgul et al., 2009; Ryding et al., 2024; Weeks et al., 2020).

The ecological hypothesis posits that climate change influences trophic interactions, thus leading to a reduction in body size.

Changes in temperature can result in temperature-dependent or -independent mechanisms that influence body size (reviewed in Gardner et al., 2011). Among temperature-independent mechanisms, decreases in food availability caused by habitat loss or fragmentation may lead to the reduction of body size (Schmidt & Jensen, 2005). Thus, limiting juveniles access to resources could lead to smaller adult body size (Rode et al., 2010; Céline Teplitsky et al., 2008). Reductions in body size have been attributed to genetic microevolutionary responses to warming; smaller individuals dissipate heat better than larger individuals due to the larger ratio between surface:volume of their bodies (*e.g.*, Bergmann's rule, Celine Teplitsky & Millien, 2014).¹

1.1 Research Question

Here, we will conduct a large scale comparative phylogenetic analysis of Neotropical passerines that inhabit the Brazilian Atlantic Forest to investigate whether temporal changes in climate variables are associated with body size changes.

¹ I think that this paragraph is not necessary. Talk only about the ecological consequences.

1.2 Hypothesis & Predictions

2 Data & Methods

2.1 Data Collection

2.2 Target Sample Sizes

2.3 Data Analysis Plan

3 Conclusion

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